2013 Q5

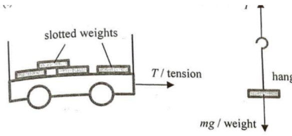
5.	Stick the paper strip onto the horizontal part BC of the track.	1A
	Release the toy from a certain height h from the bench surface and measure the corresponding stopping distance d.	1A
	Stopping distance should be measured from the beginning of the horizontal part BC or on the paper strip.	
	Release the toy from different heights and measure the corresponding stopping distances.	1A
	Plot a graph of d against h,	1A
	a straight line passing through the origin should be obtained.	1A
	or since $mgh = Fd$ so $\frac{d}{h} = \text{constant}$ / $d \propto h$	
		5

2018 Q3

3. (a)	If the maximum load is exceeded, the braking distance will increase if the friction provided remains the same.	1A	
	Vehicles would not be able to stop in time in case of emergency (thus dangerous).	1A	
	Or A larger friction is required in braking the vehicle within the same distance,	1A	
	accident may occur if the brakes cannot provide such frictional forces.	1A	
		2	
(b) (i)	If the brakes are applied continuously, thermal energy generated will heat up the brake pads / brakes to too high a temperature that the brakes may fail.	1A	
		1	
(ii)	Let D be the distance travelled along the ramp. Kinetic energy of vehicle becomes its gravitational potential energy: $\frac{1}{2}m(25)^2 = m(9.81)(D\sin 30^\circ)$	1M	Or work done against vehicle's weight component along the ramp by its kinetic energy: $\frac{1}{2}mv^2 = mg\sin\theta \times D$
	$D = 63.710499$ m ≈ 63.7 m (62.5 m for $g = 10$ m s^{-2})	1A	
	Or $v^2 = u^2 + 2as$	1M	mass of vehicle = m Note: $D\sin 30^\circ = 31.8552$ m

	Marks	Total	Remarks
3. (a) (i) Work done = K.E. gained			
$= \frac{1}{2}(0.22)(13.4)^2$	1M		
$= 19.7516\text{ J} \approx 19.8\text{ J}$	1A		
		2	
(ii) By conservation of mechanical energy,			
$\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = mgh$	1M		
$v^2 - 13.4^2 = 2(9.81)(1.0)$			
$v = 14.113114\text{ m s}^{-1} \approx 14.1\text{ m s}^{-1}$	1A		For $g = 10\text{ m s}^{-2}$, $v = 14.13\text{ m s}^{-1}$
Or			
$v_y^2 = u_y^2 + 2as$			
$v_y^2 = (13.4 \sin 55°)^2 + 2(-9.81)(-1.0)$	1M		
$v_y = 11.836662\text{ m s}^{-1} \approx 11.8\text{ m s}^{-1}$			
speed $v = \sqrt{v_x^2 + v_y^2}$			
$= \sqrt{11.8^2 + (13.4 \cos 55°)^2}$			
$= 14.113079\text{ m s}^{-1} \approx 14.1\text{ m s}^{-1}$	1A		Accept 14.1 m s^{-1} to 14.2 m s^{-1}
		2	
(b) (i) descending (a pointing downward) $-g$	1A		
		1	
(ii) $\frac{18}{2} = 13.4 \cos 55° \times t$ (constant horizontal speed)	1M		
$t = 1.170971\text{ s} \approx 1.17\text{ s}$	1A		
		2	
(c) Justified (with correct explanation)	1A		
Horizontal component of the (initial) velocity is (greatly) increased (for a similar speed but at a much smaller angle (35° << 55°))	1A		
Or			
Vertical component of the (initial) velocity is (greatly) reduced / vertical height reached is smaller.			
Time of flight (to the highest point as well as) for the whole journey would be shortened.			
		2	
(d) (Wood is soft compared to concrete, thus) it lengthens the time of impact when the feet land on the ground. This reduces the impact force.	1A 1A		
[as net force $F = \frac{mv - mu}{t} = \frac{0 - mu}{t}$]			
		2	

4. (a) (i)



	Marks	Total	Remarks
(a) (i) (forces on diagram)	1A 1A		
		2	
(ii) Smaller than (the weight of hanger / mg), i.e. $T < mg$,	1A		
therefore by Newton's second law, a net force F is provided for the downward acceleration of the hanging mass	1A		
Or			
therefore by Newton's second law, a downward net force F is provided for the acceleration of the hanging mass (both F and a are downward).			
		2	
(iii) Applying Newton's second law to the whole system, $mg = (M + 0.1)a$	1A		
		1	
(b) slope of the graph $= \frac{2.5}{0.1} = 25$ (m s^{-2} kg^{-1}) or 0.025 (m s^{-2} g^{-1})	1A		
$mg = (M + 0.1)a$			
$a = \frac{g}{M+0.1} \times m$	1M		
$\therefore$ slope of the graph $= 25 = \frac{9.81}{M+0.1}$			
$M = 0.2924\text{ kg} \approx 292\text{ g} \approx 0.3\text{ kg}$	1A		For $g = 10\text{ m s}^{-2}$, $M = 0.3\text{ kg}$
		3	

	Marks	Total	Remarks
5. (a) (i) $F = \frac{\Delta p}{\Delta t} = \frac{2.60 \times 10^3 \times v}{1} = 5.20 \times 10^6$	1M		force on gas = thrust on rocket in magnitude
$v = 2000\text{ m s}^{-1}$	1A		
		2	
(ii) $F - mg = ma$			
$a = \frac{F}{m} - g = \frac{5.2 \times 10^6}{3.6 \times 10^5} - 8.56$	1M		
$= 5.884444\text{ m s}^{-2} \approx 5.88\text{ m s}^{-2}$	1A		Accept 5.88 m s^{-2} to 5.90 m s^{-2}
		2	
(iii) The acceleration would increase. $F=ma$	1A		
Although the thrust remains the same, the mass of the rocket and/or g decreases.	1A		
		2	
(b) (i) 24 hours / 1 day / 86400 s	1A		
		1	
(ii) $m\omega^2 r = \frac{GMm}{r^2}$ OR $\frac{mv^2}{r} = \frac{GMm}{r^2}$ OR $gR^2 = v^2 r = (\frac{2\pi r}{T})^2 r$	1M		